

PROBLEM STATEMENT

Nested Square Root Differentiation

Find the derivative with respect to x of the function:

$$y = \sqrt{x^2 + \sqrt{x^2 + 1}}$$

STEP 1 প্রদত্ত দ্বিস্তরীয় বর্গমূলীয় ফাংশনটি উল্লেখ করি (Given Nested Radical Function)

$$y = \sqrt{x^2 + \sqrt{x^2 + 1}}$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} \left[\sqrt{x^2 + \sqrt{x^2 + 1}} \right]$$

STEP 3 বহিঃস্থ বর্গমূলের ওপর চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[\sqrt{u}] = \frac{1}{2\sqrt{u}} \frac{du}{dx}$)

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x^2 + \sqrt{x^2 + 1}}} \cdot \frac{d}{dx} (x^2 + \sqrt{x^2 + 1})$$

STEP 4 অভ্যন্তরীণ রাশির অন্তরক নির্ণয় ও সরলীকরণ করি (Differentiating Inner Sum)

$$\frac{d}{dx} (x^2 + \sqrt{x^2 + 1}) = 2x + \frac{2x}{2\sqrt{x^2 + 1}} = 2x + \frac{x}{\sqrt{x^2 + 1}} = \frac{x(2\sqrt{x^2 + 1} + 1)}{\sqrt{x^2 + 1}}$$

STEP 5 মানগুলো গুণ করে সম্মিলিত ভগ্নাংশ আকারে চূড়ান্ত অন্তরজ পাই (Final Algebraic Assembly)

$$\frac{dy}{dx} = \frac{x(2\sqrt{x^2 + 1} + 1)}{2\sqrt{x^2 + 1}\sqrt{x^2 + \sqrt{x^2 + 1}}}$$

FINAL ANSWER

$$\frac{dy}{dx} = \frac{x(2\sqrt{x^2 + 1} + 1)}{2\sqrt{x^2 + 1}\sqrt{x^2 + \sqrt{x^2 + 1}}}$$

✦ BOX 1: FORMULATION & NESTED RADICAL CHAIN RULES

- **Nested Chain Formula:** $\frac{d}{dx} \sqrt{u(x)} = \frac{u'(x)}{2\sqrt{u(x)}}$.
- Here, $u(x) = x^2 + \sqrt{x^2 + 1}$.
- Then $u'(x) = 2x + \frac{x}{\sqrt{x^2+1}} = x \left(2 + \frac{1}{\sqrt{x^2+1}} \right) = \frac{x(2\sqrt{x^2+1}+1)}{\sqrt{x^2+1}}$.
- Multiplying the outer denominator $2\sqrt{u(x)}$ produces the elegant composite fraction.

⚠ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Dropping Inner Radical's Chain Factor:** Writing $\frac{d}{dx} \sqrt{x^2 + 1} = \frac{1}{2\sqrt{x^2+1}}$ without multiplying by $2x$.
- **Failing to Factor Common x :** Leaving unsimplified fractions scattered without bringing them under a common denominator.

🌐 BOX 3: REAL DOMAIN & GLOBAL MINIMUM

- **Domain of Definition:** Since $x^2 + 1 \geq 1 > 0$ and $x^2 + \sqrt{x^2 + 1} \geq 1 > 0$ for all $x \in \mathbb{R}$, the function is defined and smooth on all of $\mathbb{R} = (-\infty, \infty)$.
- **Stationary Point:** Setting $\frac{dy}{dx} = 0 \iff x(2\sqrt{x^2 + 1} + 1) = 0$. Since $2\sqrt{x^2 + 1} + 1 \geq 3 > 0$, the only stationary point is $x = 0$.
- At $x = 0$, $y = \sqrt{0 + \sqrt{1}} = 1$ (global minimum of the curve).

💡 BOX 4: LEIBNIZ SUBSTITUTION (ALTERNATIVE VERIFICATION)

- Let $v = \sqrt{x^2 + 1} \implies \frac{dv}{dx} = \frac{x}{\sqrt{x^2+1}}$.
- Let $u = x^2 + v \implies \frac{du}{dx} = 2x + \frac{dv}{dx} = 2x + \frac{x}{\sqrt{x^2+1}}$.
- Let $y = \sqrt{u} \implies \frac{dy}{du} = \frac{1}{2\sqrt{u}}$.
- By Chain Rule:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{2\sqrt{u}} \left[\frac{x(2\sqrt{x^2+1}+1)}{\sqrt{x^2+1}} \right] = \frac{x(2\sqrt{x^2+1}+1)}{2\sqrt{x^2+1}\sqrt{x^2+\sqrt{x^2+1}}}$$

📊 BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 4 / 5 MARKS)

- **1.0 Mark:** Correct outer radical chain rule setup $\frac{1}{2\sqrt{x^2+\sqrt{x^2+1}}}$.
- **2.0 Marks:** Accurate step-by-step differentiation of inner radical $2x + \frac{x}{\sqrt{x^2+1}}$.
- **1.0 Mark:** Common denominator simplification and final boxed presentation.